GAUTAM SINGH

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EDUCATION

Conestoga College — Guelph, ON
Bachelor of Computer Science (Honors) — *Sep 2020 - Dec 2025*
Relevant coursework: Machine Learning, Data Structures & Algorithms, Database Design, Cloud Computing (Azure, Databricks), Software Engineering, Object-Oriented Programming

Google Advanced Data Analytics Professional Certificate — *Jun 2026*
Google / Coursera — *Online*

EXPERIENCE

ServiceNow Developer & QA Specialist — Sep 2024 - Dec 2024, Sep 2025 - Dec 2025
PAMT Consulting (now Ateko, a Bell Canada company) — *Remote*
Supported enterprise clients across two co-op terms, building ServiceNow workflows and integrations and validating them before release.

- Reduced onboarding handling time by roughly 30% with an automated workflow built in ServiceNow App Engine Studio and Flow Designer. Staff had processed each new hire by hand, which was slow and error prone.
- Integrated separate enterprise systems through ServiceNow IntegrationHub. The integration moves data between platforms automatically and removed a recurring source of transcription errors.
- Designed and executed test cases across development, test, and production environments. These checks confirmed data accuracy and integration reliability before each release and caught defects internally, before end users hit them.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, TypeScript, Java, C++, C, Rust
Frameworks & Web: React, HTML, CSS, FastAPI, Django, REST APIs, Streamlit, Bootstrap, Flutter
Data & Cloud: Azure Data Factory, Databricks, PySpark, Spark, Azure, AWS, Docker, CI/CD, Microservices, dbt, Prefect
Databases: PostgreSQL, Azure SQL, MySQL, Supabase, MongoDB, Redis, RDBMS
ML & Tools: Pandas, NumPy, scikit-learn, PyTorch, XGBoost, LangChain, LLMs, Git, Version Control, ServiceNow, Power BI, Grafana

PROJECTS

Cairn — Live | GitHub | *2026*
Rust, AWS Lambda, DynamoDB, CloudFront, API Gateway, Terraform

- Served analytics events in 5.1 ms warm behind a 184 ms cold start by writing the ingest path as a Rust binary on ARM64 AWS Lambda, doing a single DynamoDB write per event and deriving every dimension later so a visitor waits on nothing but the write.
- Deployed the full stack of Lambda, DynamoDB, CloudFront, and API Gateway as 41 Terraform-managed resources running live across three sites at zero steady-state cost, and made the service cookie-free by hashing visitor identifiers under a key that rotates daily so no address is ever stored.

Fitmit — Live | GitHub
Python, FastAPI, React, TypeScript, Redis, Docker, Ghostscript, Pillow

- Cut worst-case compression attempts from twelve to four with a binary search over the quality ladder, and shipped the service live on Render and Vercel.
- Extended the compressor to five image formats with no change to the existing search or job queue by abstracting format specific work behind one interface.

EDA Web Service — Live | GitHub
Python, FastAPI, React, JavaScript, Supabase, Plotly, Vercel, Render

- Developed a full-stack web application that automates exploratory data analysis and replaces the slow work of judging dataset quality by hand. Users upload a CSV and the app returns column type inference, data quality flags, and a chart for every column.
- Improved chart load speed threefold by cutting the Plotly bundle from 5.1 MB to 1.8 MB. Implemented report storage in Supabase with shareable links that expire after five days, and added validation limiting uploads to safe file types and sizes.

The Almanac (Capstone) — GitHub | *2025*
Python, Azure Data Factory, Databricks, PySpark, Azure SQL, PyTorch, Django REST, Flutter

- Increased data load performance by roughly 35% by redesigning the storage layer of the pipeline behind a stock tracking web and mobile application that my team shipped over eight months. The pipeline ingests, cleans, and validates market data without manual steps.
- Trained a PyTorch forecasting model to 60 to 70% next-day price accuracy on several years of historical prices. A single Django REST API serves those predictions to the web client and the Flutter mobile app, so both read from one source of truth.

Financial Document Q&A (RAG) — GitHub | *2026*
Python, LangChain, FAISS, FastAPI, Streamlit, Docker

- Built a retrieval-based question answering system over SEC filings using LangChain and FAISS for hybrid retrieval, a FastAPI backend serving queries through a Streamlit interface, and Docker for consistent deployment.